

Beijing-Dublin International College (BDIC)

SECOND SEMESTER, Academic Year 2014–2015

Campus: Beijing University of Technology (BJUT)

Advanced Mathematics (Module 4; Engineering) — Final Exam

Honesty Pledge:

I have read and clearly understand the Examination Rules of Beijing University of Technology and University College Dublin and am aware of the Punishment for Violating the Rules of Beijing University of Technology and University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I would accept the punishment thereof.

Pledger: _____

Class NO: _____

BJUT Student ID: _____

UCD Student ID: _____

NOTE: Answer **ALL** questions.

Time allowed is **90** minutes.

The exam paper has **2** sections on **8** pages, with a full score of 100 marks.

You are required to use only the provided **Examination Book** for answers.

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SECTION A — MULTIPLE CHOICE QUESTIONS

In each question, choose **at most one** option.

Circle the preferred choice on the **Examination Book** provided.

This section is worth a total of **60** marks, with each question worth **4** marks.

1. Determine the shape of the surface given by the implicit function

$$F(x, y, z) = x^2 + 2y^2 + 3z^2 - 4x + 8y - 6z = 0.$$

- (a) A sphere, centered at $(2, -2, 1)$, with radius 3;
- (b) A sphere, centered at $(2, -2, 1)$, with radius 2;
- (c) A sphere, centered at $(-2, 2, -1)$, with radius 3;
- (d) An ellipsoid, with the three half-axes being $a = \sqrt{15}$, $b = \frac{\sqrt{30}}{2}$ and $c = \sqrt{5}$.

2. Evaluate the limit

$$\lim_{(x,y) \rightarrow (0,4)} \frac{\sqrt[7]{xy+1} - 1}{x}.$$

- (a) 2;
- (b) 0;
- (c) $\frac{4}{7}$;
- (d) The limit does not exist.

3. Determine the shape of the surface described by the equation

$$x^2 + y^2 = 3z^2.$$

- (a) An ellipsoid;
- (b) A paraboloid;
- (c) A cone;
- (d) A cylindrical surface.

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4. Find the total differential of the following two-variable function

$$z = x^5 \sin y - y^2 e^x.$$

(a) $dz = (5x^4 \sin y - y^2 e^x) dx + (x^5 \cos y - 2ye^x) dy;$

(b) $dz = (5x^4 \sin y - y^2 e^x) + (x^5 \cos y - 2ye^x);$

(c) $dz = (5x^4 \sin y - y^2 e^x) dx - (x^5 \cos y - 2ye^x) dy;$

(d) $dz = (5x^4 \sin y) dx + x^5 \cos y dy - e^x (y^2 dx - 2y dy).$

5. Consider a function

$$z = f(3x - y^3, y \sin x),$$

where f has continuous partial derivatives. Then $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are respectively

(a) $\frac{\partial z}{\partial x} = f_1 \cdot 3 - f_2 \cdot y \cos x, \quad \frac{\partial z}{\partial y} = f_1 \cdot (-3y^2) + f_2 \cdot \sin x;$

(b) $\frac{\partial z}{\partial x} = f_1 \cdot 3 + f_2 \cdot y \cos x, \quad \frac{\partial z}{\partial y} = f_1 \cdot (-3y^2) + f_2 \cdot \sin x;$

(c) $\frac{\partial z}{\partial x} = f_1 \cdot 3 + f_2 \cdot y \cos x, \quad \frac{\partial z}{\partial y} = f_1 \cdot \left(\frac{y^4}{4}\right) + f_2 \cdot \sin x;$

(d) $\frac{\partial z}{\partial x} = f_1 \cdot 3 + f_2 \cdot y \cos x, \quad \frac{\partial z}{\partial y} = f_1 \cdot (-3y^2) + f_2 \cdot \cos x.$

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6. Consider a function

$$z = f(ax + by, cx + dy),$$

where a, b, c and d are constants, and f has continuous second order partial derivatives. Then $\frac{\partial^2 z}{\partial x \partial y}$ can be expressed by

(a) $\frac{\partial^2 z}{\partial x \partial y} = abf_{11} + (ac + bd)f_{12} + cdf_{22};$

(b) $\frac{\partial^2 z}{\partial x \partial y} = abf_{11} + (ad + bc)f_{12} + cdf_{22};$

(c) $\frac{\partial^2 z}{\partial x \partial y} = a^2 f_{11} + (ad + bc)f_{12} + d^2 f_{22};$

(d) $\frac{\partial^2 z}{\partial x \partial y} = abf_{11} + (ad - bc)f_{12} + cdf_{22}.$

7. Let Σ be a plane in the 3-dimensional space,

$$\Sigma : \quad x - 2y + 3z = 10.$$

Let $P(1, 2, 3)$ be a point outside the plane Σ . Find the distance between P and the plane Σ .

(a) $\frac{\sqrt{14}}{7};$

(b) $\frac{2\sqrt{14}}{7};$

(c) 4;

(d) 1.

8. Suppose we are about to use 24m^2 of cardboard to make a rectangular box without a lid. Let x, y, z (in meters) be the lengths of three adjacent edges of this box, respectively. In order to find the maximum value of the volume of the box, we can appeal to the method of Lagrange Multiplier. Then the auxiliary function we need to construct is

(a) $L(x, y, z; \lambda) = xyz + \lambda(xy + yz + zx - 12);$

(b) $L(x, y, z; \lambda) = xyz + \lambda(24 - 2xy - 2yz - 2zx);$

(c) $L(x, y, z; \lambda) = xyz + \lambda(12 - xy - yz - zx);$

(d) $L(x, y, z; \lambda) = xyz + \lambda(24 - xy - 2yz - 2zx).$

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9. Let Σ be a surface described by the equation

$$x^2 - 9y^2 + \frac{z^2}{4} = 1.$$

The tangent plane to Σ at the point $P(3, 1, -2)$ is given by

- (a) $6x - 18y - z + 2 = 0$; (b) $3x - 9y - \frac{1}{2}z = 1$;
 (c) $6x - 18y + z + 2 = 0$; (d) $6x - 8y + z + 2 = 0$.

10. Let C be a curve in space

$$C : \begin{cases} x = \cos 3t, \\ y = \sin 3t, \\ z = t^2. \end{cases}$$

Find the tangent line to C at the point where $t = \frac{\pi}{4}$.

- (a) $\frac{x - \frac{\sqrt{2}}{2}}{-\frac{3}{\sqrt{2}}} = \frac{y - \frac{\sqrt{2}}{2}}{-\frac{3}{\sqrt{2}}} = \frac{z - \frac{\pi^2}{16}}{\frac{\pi}{2}}$; (b) $\frac{x - \frac{\sqrt{2}}{2}}{-\frac{3}{\sqrt{2}}} = \frac{y + \frac{\sqrt{2}}{2}}{-\frac{3}{\sqrt{2}}} = \frac{z - \frac{\pi^2}{16}}{\frac{\pi}{2}}$;
 (c) $\frac{x + \frac{\sqrt{2}}{2}}{-\frac{3}{\sqrt{2}}} = \frac{y - \frac{\sqrt{2}}{2}}{-\frac{3}{\sqrt{2}}} = \frac{z - \frac{\pi^2}{16}}{\frac{\pi}{2}}$; (d) $\frac{x + \frac{\sqrt{2}}{2}}{-\frac{3}{\sqrt{2}}} = \frac{y - \frac{\sqrt{2}}{2}}{\frac{3}{\sqrt{2}}} = \frac{z - \frac{\pi^2}{16}}{\frac{\pi}{2}}$.

11. Consider a surface described by the function $z = f(x, y) = xy$, and a point $P(0, 0)$ on the surface.

Which of the following statements is correct?

- (a) P is a stationary point and a minimum point of $f(x, y)$;
 (b) P is a stationary point and a maximum point of $f(x, y)$;
 (c) P is a stationary point but not an extreme point of $f(x, y)$;
 (d) P is neither a stationary point nor an extreme point of $f(x, y)$.

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12. Let $f(x, y) = x^2y^3 - y^5$. At the point $P(1, 1)$, the directional derivative of $f(x, y)$, denoted as $D_{\mathbf{v}}f$, along the vector $\mathbf{v} = (1, 1)$ is given by

- (a) $-26\sqrt{2}$; (b) 0; (c) $4\sqrt{2}$; (d) $\sqrt{2}$.

13. Let $f(x, y) = e^{x^2y} + \sin 2y$. The gradient of f is given by

- (a) $\nabla f = 2xe^{x^2y}\mathbf{i} + (x^2e^{x^2y} + 2\cos 2y)\mathbf{j}$;
 (b) $\nabla f = e^{x^2y}\mathbf{i} + (x^2e^{x^2y} + 2\cos 2y)\mathbf{j}$;
 (c) $\nabla f = 2xe^{x^2y}\mathbf{i} + (x^2e^{x^2y} + \cos 2y)\mathbf{j}$;
 (d) $\nabla f = 2xe^{x^2y} + (x^2e^{x^2y} + 2\cos 2y)$.

14. Evaluate the integral

$$\iint_{D: x^2+y^2 \leq 1} (x^3 \cos y - yx^2 - 5) dx dy.$$

- (a) 0; (b) 5; (c) -5 ; (d) -5π .

15. Consider a double integral

$$I = \iint_D xy dx dy, \quad \text{with } D = \{(x, y) | 0 \leq x \leq 1; x \leq y \leq 1\}.$$

It can be converted to polar coordinates as

- (a) $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{\sec \theta} r \cos \theta \cdot r \sin \theta \cdot r dr$;
 (b) $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{\csc \theta} r \cos \theta \cdot r \sin \theta \cdot r dr$;
 (c) $I = \int_0^{\frac{\pi}{4}} d\theta \int_0^{\csc \theta} r \cos \theta \cdot r \sin \theta \cdot r dr$;
 (d) $I = \int_0^{\frac{\pi}{4}} d\theta \int_0^{\sec \theta} r \cos \theta \cdot r \sin \theta \cdot r dr$.

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SECTION B — EXTENDED ANSWER QUESTIONS

Write your answers on the **Examination Book** provided.

This section is worth a total of **40** marks. The marks of each question are as shown.

- 16. (7 marks)** Compute the following iterated integral by converting it to polar coordinates

$$I = \int_{-2}^2 dx \int_0^{\sqrt{4-x^2}} \sin(x^2 + y^2) dy.$$

- 17. (7 marks)** Evaluate the following triple integral

$$I = \iiint_{\Omega} (z + xz^2 + yz) dv,$$

where Ω is a region bounded by the plane $z = 2$ and the cone $z = \sqrt{x^2 + y^2}$.

- 18. (7 marks)** Evaluate the iterated integral

$$I = \int_1^2 dx \int_{\sqrt{x}}^x \sin \frac{\pi x}{2y} dy + \int_2^4 dx \int_{\sqrt{x}}^2 \sin \frac{\pi x}{2y} dy.$$

- 19. (7 marks)** Express the following triple integral as an iterated integral in spherical coordinates:

$$I = \iiint_{\Omega} f(x, y, z) dv$$

where Ω is a region bounded by the cone $z = \sqrt{x^2 + y^2}$ and the hemisphere $z = \sqrt{2 - x^2 - y^2}$.

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20. (7 marks) Let Ω be a 3 dimensional solid body bounded by the surfaces

$$z = x^2 + y^2 \quad \text{and} \quad z = \sqrt{6 - (x^2 + y^2)}.$$

Compute the volume of Ω .

21. (5 marks) Prove

$$\int_a^b dx \int_a^x (x - y)^3 f(y) dy = \frac{1}{4} \int_a^b (b - y)^4 f(y) dy,$$

where $f(y)$ is continuous over the interval $[a, b]$.